

The image features the text "DL for ENG" in a bold, 3D, gold-colored font. The letters are floating on a vast, blue ocean with visible ripples and small waves. In the background, there are rolling blue hills or mountains under a sky filled with soft, grey clouds. The overall scene has a serene and expansive feel.

DL for ENG

LECTURE 10.1

Ionic Diffusion & Charge Conservation

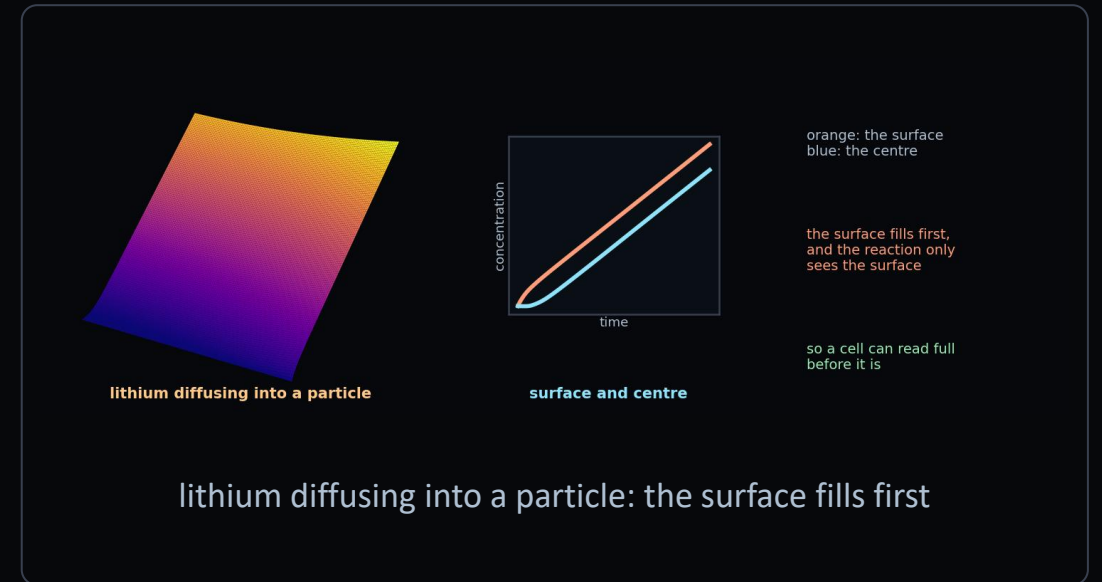
Battery models with PINNs: from a single particle to electrolyte transport and thermal coupling

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Aalborg University · ret@et.aau.dk · Deep Learning for Engineering · 2026

Doyle & Newman DFN · Gu & Wang (2000) · PyBaMM: Chen2020, O'Regan2022 (LG M50 21700)



What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **name** — the five fields in a cell model, and the two length scales they live on
- **choose** — between SPM, SPM_e and DFN, and defend the choice by C-rate
- **explain** — why the surface concentration matters and the average does not
- **scale** — an electrochemical problem, where raw SI units span twenty orders
- **argue** — why thermal and electrochemical coupling cannot be separated
- **identify** — a diffusivity from a voltage curve, and say when it is not identifiable

WHAT TODAY ASSUMES

- | | |
|-----------|--------------------------------------|
| from L8.2 | the parabolic operator, and time |
| from L8.1 | elliptic charge equations |
| from L8.2 | identifying a parameter from a curve |

THE SAME EQUATIONS, NEW NAMES

Solid diffusion is L8.2 in spherical coordinates.
Charge conservation is L8.1's Poisson problem.
What is new is that they are coupled.

AND ONE SEVERE NONLINEARITY

Butler-Volmer is an exponential. It is the harshest term in the course.

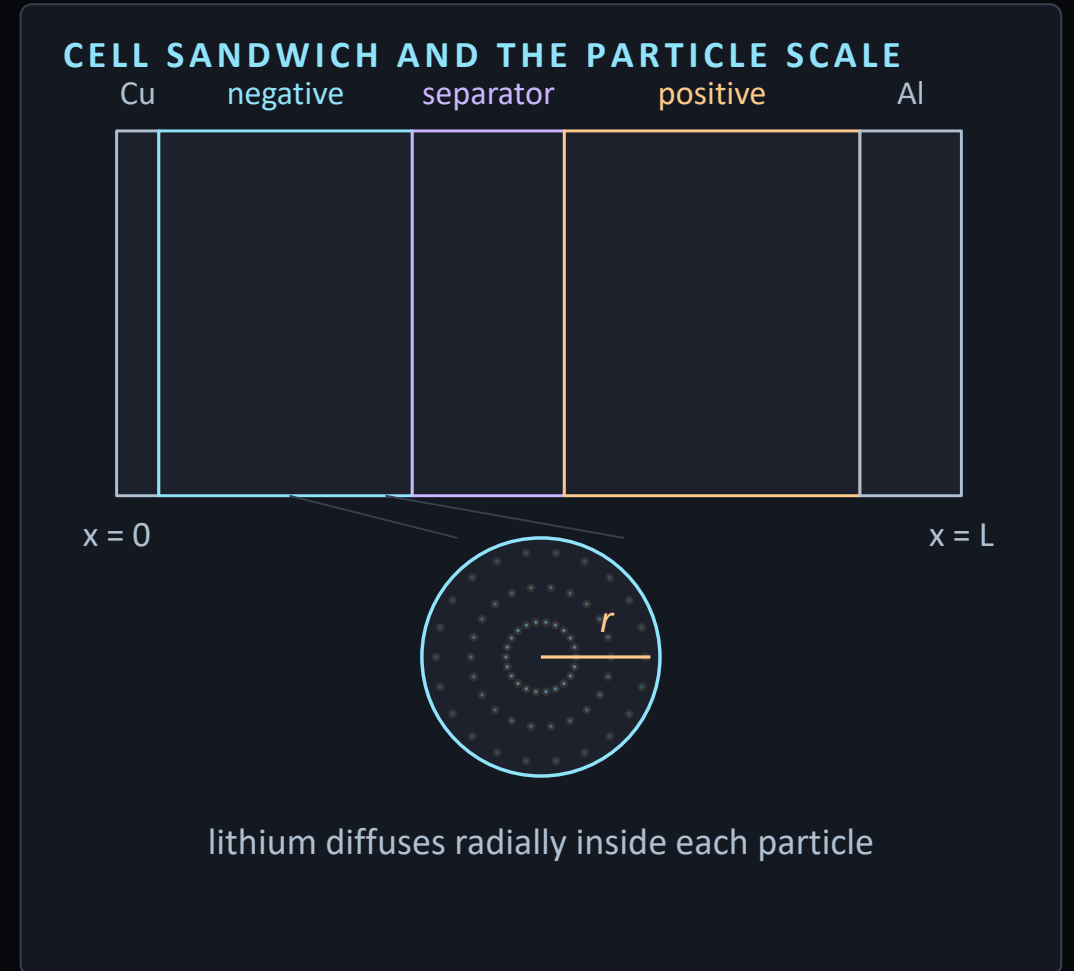
The First Genuinely Coupled Multiphysics Problem

	L9 — FLOW	L10 — ELECTROCHEMICAL
Unknowns	u, v, p	$c_s, c_e, \varphi_s, \varphi_e, T$
Physics	<i>one — momentum</i>	<i>three — transport, charge, heat</i>
Coupling	<i>within one system</i>	<i>each physics changes the others</i>
Nonlinearity	<i>convection</i>	<i>Butler–Volmer, exponential in η</i>
Dimensions	x, y, t	x, t and a particle radius r
Scales	<i>one</i>	μm particles, cm cell, hours of discharge

Every previous lecture solved one physics. This one couples three, and the coupling runs both ways — which is exactly why Gu & Wang argue a decoupled model gets the answer wrong.

Inside the Cell: Five Fields, Two Scales

- **a cell is a sandwich** — collectors, two porous electrodes, a separator
- both electrodes are porous solids flooded with electrolyte
- so lithium lives in two places: inside particles, and dissolved between them
- that gives four electrochemical fields, plus temperature
- **and a second scale** — the particle, where lithium must diffuse to reach the surface



SPM, SPM_e, DFN — Choose Deliberately

SINGLE PARTICLE MODEL

SPM

One particle per electrode.
Electrolyte assumed uniform.
Two PDEs, both radial diffusion.
Accurate below about 1C.
Where this lecture starts.

SPM WITH ELECTROLYTE

SPM_e

Adds electrolyte concentration
and potential along x.
Four fields, still no distributed
reaction rate.
Good to roughly 2–3C.

DOYLE–FULLER–NEWMAN

DFN

A particle at every point in x.
Five coupled fields, full
Butler–Volmer distribution.
Accurate at high rate.
Research-level for a PINN.

- the hierarchy is an engineering choice, not a ladder of sophistication
- at low C-rate the simplest model is adequate, and for a PINN the choice matters more
- each added field is another network, another residual, another thing to balance

Diffusion Inside a Particle

RADIAL DIFFUSION IN A SPHERICAL PARTICLE

$$\frac{\partial c_s}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 D_s \frac{\partial c_s}{\partial r} \right)$$

c_s the lithium concentration in the particle, r the radius, D_s the solid diffusivity

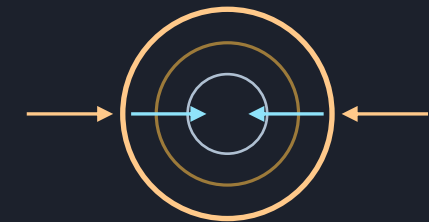
SYMMETRY AT THE CENTRE, REACTION FLUX AT THE SURFACE

$$\left. \frac{\partial c_s}{\partial r} \right|_{r=0} = 0, \quad -D_s \left. \frac{\partial c_s}{\partial r} \right|_{r=R_s} = \frac{j}{a_s F}$$

R_s the particle radius, j the reaction current per volume
 a_s the surface area per volume, F Faraday's constant

- this is L8.2's heat equation in spherical coordinates — the same parabolic operator
- the electrochemistry enters at the surface, through the reaction flux
- and it is the surface concentration, not the average, that the reaction sees
- at high rate the surface saturates long before the centre does
- which is why a cell can appear full while most of the lithium has not arrived

A PARTICLE, DRAWN



lithium enters at the surface
and diffuses to the centre

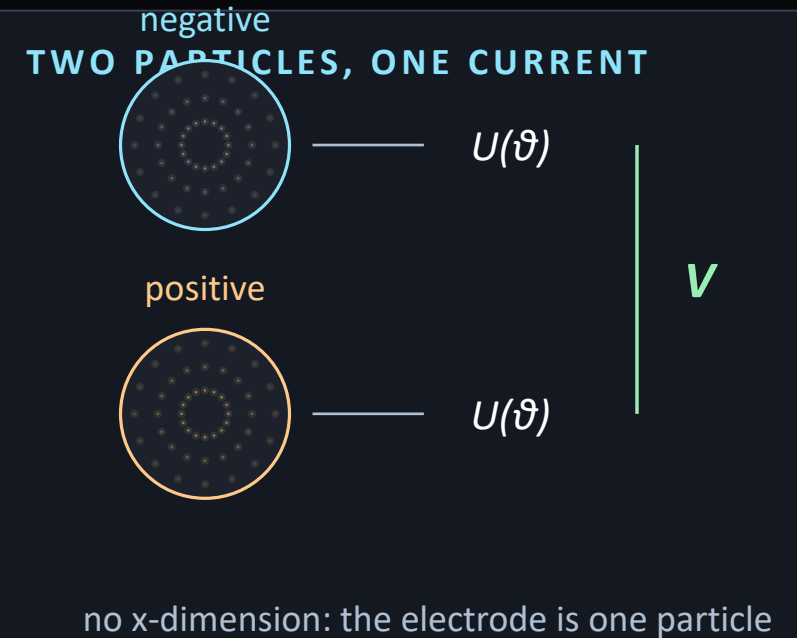
The Single Particle Model

CELL VOLTAGE FROM TWO PARTICLES

$$V = U_p(\theta_p) - U_n(\theta_n) - \eta_p - \eta_n$$

U_p, U_n the open circuit potentials, θ the surface state of charge
 η_p, η_n the overpotentials of the two electrodes

- **one assumption does all the work** — a uniform reaction rate through each electrode
- the model reduces to two radial diffusion problems, coupled only through the current
- the voltage is the difference of two open-circuit potentials, minus the overpotentials
- for a PINN this is an excellent starting point
- two one-dimensional parabolic problems, and one algebraic relation



Butler–Volmer: the Reaction at the Interface

CHARGE-TRANSFER KINETICS

$$i_n = i_0 \left[\exp \left(\frac{\alpha_a F \eta}{RT} \right) - \exp \left(-\frac{\alpha_c F \eta}{RT} \right) \right]$$

i_n the reaction current density, i_0 the exchange current density, η the overpotential
 α_a, α_c the transfer coefficients, F Faraday's constant, R the gas constant
 $c_e, c_{se}, c_{s, \max}$ electrolyte, surface and maximum concentration, k_r the rate constant

SURFACE OVERPOTENTIAL

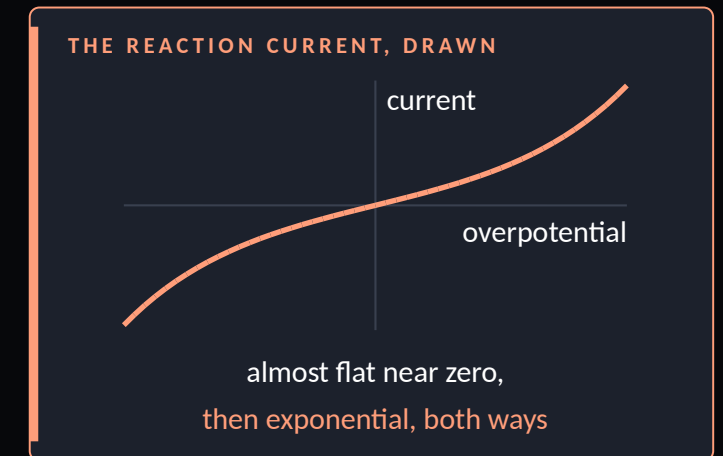
$$\eta = \phi_s - \phi_e - U(\theta, T)$$

ϕ_s, ϕ_e the potentials of solid and electrolyte, U the open circuit potential

EXCHANGE CURRENT DENSITY

$$i_0 = k_r c_e^{\alpha_a} (c_{s, \max} - c_{se})^{\alpha_a} c_{se}^{\alpha_c}$$

- two exponentials of the overpotential, one forward and one backward
- this is the most severe nonlinearity in the course
- an exponential amplifies any error in the argument
- **note where temperature enters** — inside the exponential, and again in the potential
- so train on overpotential in units of RT/F , not volts



Open-Circuit Potential and State of Charge

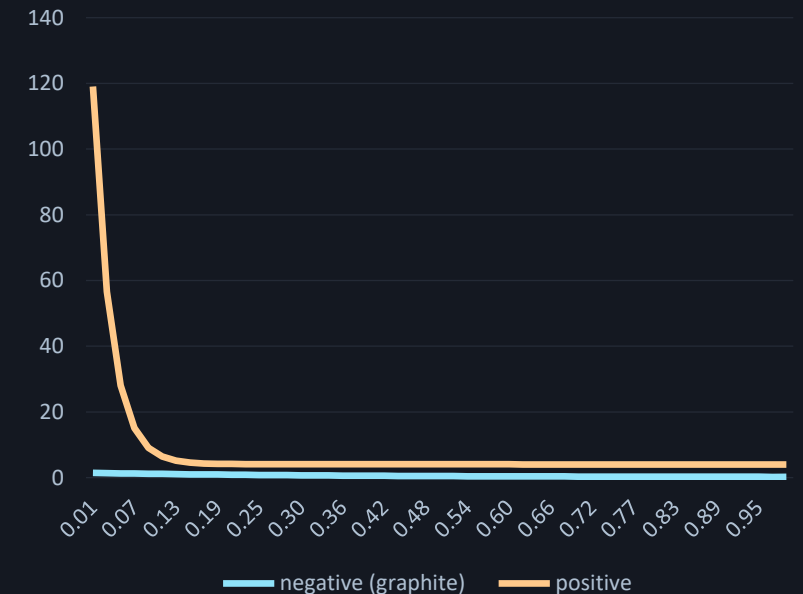
LOCAL STATE OF CHARGE

$$\theta = \frac{c_{se}}{c_{s,\max}}$$

c_{se} the surface concentration
 $c_{s,\max}$ the most a particle holds

- the open-circuit potential is the voltage the electrode would show at rest
- it is measured, not derived** — supplied as a fitted curve
- strongly nonlinear, with plateaus where a phase transition holds it flat
- the steep ends are where the model is hardest and the cell most vulnerable
- and the temperature dependence is the entropic coefficient

TYPICAL OCP CURVES



*Plateaus in the middle, steep walls at both ends.
 The steep regions set the usable window and
 are where models break.*

Adding the Electrolyte

LITHIUM CONSERVATION IN THE ELECTROLYTE

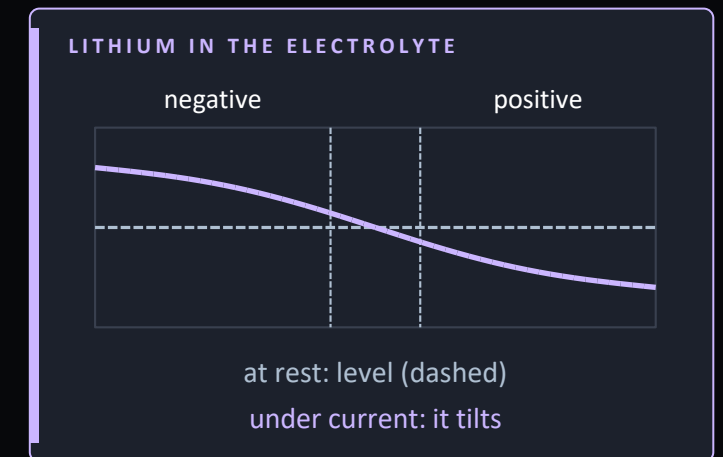
$$\varepsilon_e \frac{\partial c_e}{\partial t} = \nabla \cdot (D_e^{eff} \nabla c_e) + \frac{1 - t_+^0}{F} j^{Li}$$

c_e the lithium in the electrolyte, ε_e the porosity, D_e^{eff} the effective diffusivity
 t_+^0 the transference number, j^{Li} the reaction current per volume

BRUGGEMAN CORRECTION FOR TORTUOSITY

$$D_e^{eff} = D_e \varepsilon_e^{1.5}, \quad \kappa^{eff} = \kappa \varepsilon_e^{1.5}$$

- a **parabolic transport equation** — L8.2's heat equation again, with a source
- the source is the reaction: lithium inserted here is lithium removed there
- **the effective diffusivity is not the bulk value** — the path is tortuous
- at high rate the electrolyte depletes at one electrode and accumulates at the other
- that gradient is a real contribution to the voltage loss



Two Phases, Two Potentials

CHARGE IN THE ELECTROLYTE — NOTE THE CONCENTRATION-DRIVEN TERM

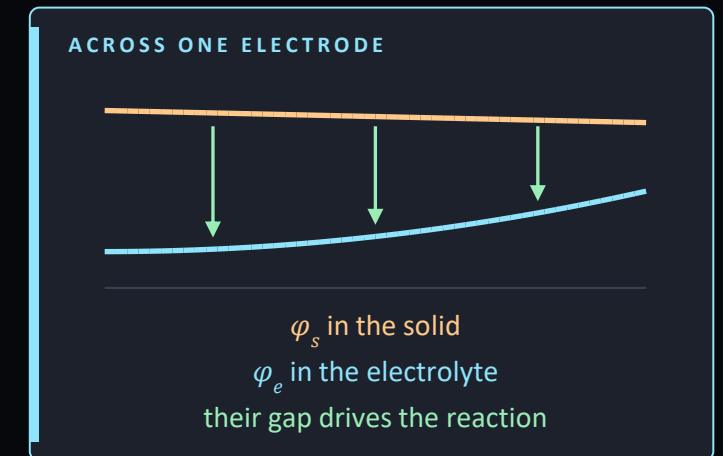
$$\nabla \cdot (\kappa^{eff} \nabla \phi_e) + \nabla \cdot (\kappa_D^{eff} \nabla \ln c_e) + j^{Li} = 0$$

ϕ_e, ϕ_s the potentials of electrolyte and solid
 $\kappa^{eff}, \sigma^{eff}$ their effective conductivities

CHARGE IN THE SOLID PHASE

$$\nabla \cdot (\sigma^{eff} \nabla \phi_s) - j^{Li} = 0$$

- **neither charge equation has a time derivative** — both are elliptic
- the Poisson problems of L7.2 and L8.1, in a new setting
- the reaction current appears in both, with opposite signs
- the second term is a diffusion potential, driven by the concentration gradient
- so a full model couples parabolic transport to elliptic charge, at every instant



Five Equations, Five Unknowns

FIELD	MEANING	TYPE	WHAT IT IS
C_s	lithium in the solid	parabolic, radial	Fick in a sphere
C_e	lithium in the electrolyte	parabolic, along x	with a reaction source
φ_s	solid-phase potential	elliptic	Ohm in the matrix
φ_e	electrolyte potential	elliptic	Ohm plus diffusion potential
T	temperature	parabolic	heat equation with a source

Every one of these five is a PDE you have already solved in isolation. What is new in L10 is solving them together, with each one changing the coefficients of the others.

The Energy Equation and Its Source

ENERGY CONSERVATION — THE L8.2 EQUATION, WITH AN ELECTROCHEMICAL SOURCE

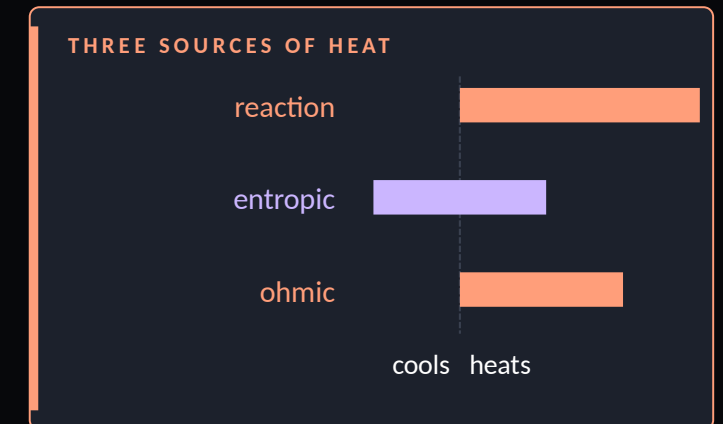
$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + q$$

ρc_p the heat capacity per volume, k the thermal conductivity, q the heat source

HEAT GENERATION: REACTION, ENTROPIC, AND OHMIC IN BOTH PHASES

$$q = a_s i_n \eta + a_s i_n T \frac{\partial U}{\partial T} + \sigma_{eff} \nabla \phi_s \cdot \nabla \phi_s + \kappa_{eff} \nabla \phi_e \cdot \nabla \phi_e$$

- three physically distinct sources
- irreversible reaction heat, always positive
- reversible entropic heat, which changes sign with the direction of current
- and ohmic heating in both phases
- **Gu and Wang neglect the entropic term, and say why** — it was not measured then



The Loop Runs Both Ways

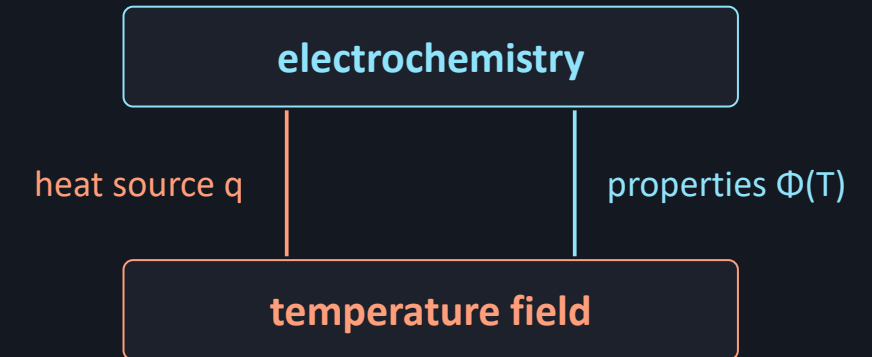
EVERY PROPERTY IS TEMPERATURE-DEPENDENT

$$\Phi = \Phi_{ref} \exp \left[\frac{E_{act, \Phi}}{R} \left(\frac{1}{T_{ref}} - \frac{1}{T} \right) \right]$$

- electrochemistry generates heat, heat raises temperature, temperature changes transport
- so the loop closes, and Gu and Wang test it directly
- a decoupled model held at reference temperature gets the trend backwards
- the physical reason is that warming a cell helps it
- **so coupling is not a refinement** — decoupling gives qualitatively wrong answers

Φ any rate constant or diffusivity
 E_{act} its activation energy, R the gas constant
 T_{ref} the temperature Φ_{ref} was measured at

THE TWO-WAY COUPLING



Drop either arrow and the model is decoupled — and, per Gu & Wang, wrong in both magnitude and trend.

Why the Temperature Field Is Not Uniform

CONVECTIVE COOLING AT THE SURFACE

$$-k \frac{\partial T}{\partial n} = h(T - T_a)$$

k the thermal conductivity
 h the heat transfer coefficient
 T_a the ambient temperature

BIOT NUMBER • LUMPED OR NOT

$$Bi = \frac{hR}{k}$$

R the radius of the cell

- Gu and Wang report up to 40 °C variation within a single large cell
- a 21700 is smaller, but the mechanism is identical
- heat is generated throughout the jellyroll and leaves only through the surface
- so the natural second dimension is the r-z plane
- **check the Biot number first** — if it is small, a lumped model is defensible

CORE HOTTER THAN CAN • TAB ON TOP



Building the Model, Field by Field

- one network per field, or one network with several outputs
- the radial particle coordinate is a genuine input
- the surface flux condition links the diffusion problem to the reaction
- **and train against a reference** — PyBaMM solves these quickly and reliably
- so the exercise is reproduction, not discovery

WHAT EACH RESIDUAL NEEDS

$c_s(r, t)$	parabolic	$\partial/\partial t, \partial^2/\partial r^2$
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$c_e(x, t)$	parabolic	$\partial/\partial t, \partial^2/\partial x^2, \text{source}$
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$\varphi_e(x)$	elliptic	$\partial^2/\partial x^2, \nabla \ln c_e$
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$T(r, z, t)$	parabolic	$\partial/\partial t, \nabla^2, \text{source } q$
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η	algebraic	no derivatives — but exponential
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the last row is the hard one: no derivatives, but an exponential

Scaling: Worse Here Than Anywhere Else

- electrochemical quantities span an extraordinary range in SI units
- fed raw, the residuals differ by twenty orders of magnitude between terms
- no weighting recovers from that
- scale concentration by its maximum, radius by the particle radius, time by the discharge
- PyBaMM does this internally, which is the pattern to follow

AN LG M50 IN SI UNITS

QUANTITY	SI	SCALED
particle radius	$\sim 5 \times 10^{-6} \text{ m}$	1
solid diffusivity	$\sim 1 \times 10^{-14} \text{ m}^2/\text{s}$	1
$C_{s,\text{max}}$	$\sim 3 \times 10^4 \text{ mol/m}^3$	1
discharge time	$\sim 3600 \text{ s}$	1
overpotential	$\sim 0.05 \text{ V}$	$\eta F/RT \sim 2$
capacity	5 Ah	—

Twenty orders of magnitude between the smallest and the largest. Scaling decides whether it trains.

PyBaMM as the Reference

- the **LG M50** is the standard public case — a 5 Ah 21700 cell
- Chen2020 is complete for isothermal work but carries no thermal parameters
- O'Regan2022 measured them, and ships a worked thermal example
- use PyBaMM for the reference solution and the parameters
- and the PINN to reproduce it, then to do what PyBaMM cannot

CHOOSING A PARAMETER SET

Chen2020

LG M50, original

isothermal work

O'Regan2022

measured thermal data

the thermal half

OKane2022

superset, degradation

ageing studies

Marquis2019

Kokam pouch cell

not cylindrical

all four are public and citable

What You Can Recover From a Voltage Curve

DIFFUSIVITY AS A TRAINABLE PARAMETER

D_s trainable, $\mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_V$ D_s the solid diffusivity, trained with the weights; $\mathcal{L}_V$ the misfit to the measured voltage

- a battery management system measures voltage, current and surface temperature
- everything else must be inferred
- that is an inverse problem of exactly the kind L8.2 introduced
- the solid diffusivity ages most and is hardest to measure directly
- **identifiability is the catch** — a relaxation period carries more information than a steady discharge

MEASURED VERSUS INFERRED

MEASURED

$V(t), I(t), T_{\text{sur}}^f$

IMPOSED

the model residuals

INFERRED

internal $c_s, c_e, T^{\text{core}}$

INFERRED

D_s , and hence state of health

the core temperature is never measured

Pitfalls Specific to Electrochemical Models

Unscaled units

Twenty orders of magnitude between raw quantities in one loss.

Symptom: loss barely moves. Fix: non-dimensionalise everything, including η .

Decoupled by accident

Properties left at their reference temperature while the thermal model runs.

Symptom: overpredicted temperature, underpredicted voltage — Gu & Wang, Fig. 2.

Butler–Volmer overflow

An exponential of a large argument during early training produces inf or nan.

Fix: clamp η , or use the linearised form until the model is near a solution.

OCP endpoints

$U(\theta)$ is nearly vertical as θ approaches 0 or 1.

Symptom: divergence at end of discharge. Fix: restrict the θ window, or sample it densely.

Wrong model chosen

An SPM used at 3C, where electrolyte depletion dominates.

Symptom: voltage error growing with rate. Fix: SPMe — not more training.

The last is a modelling error, not a numerical one — and it is the one students most often try to fix by training longer.

Questions

1 – 2 · CELL MODEL

1 What does a lithium-ion cell model solve for, and on which two length scales?

Five fields: the lithium concentration in the particles and in the electrolyte, the potential in each of the two phases, and the temperature. They live on the thickness of the electrode and on the radius of a particle, and the reaction sees the concentration at the particle surface, not its average.

Ex_10.1 sections 1 – 5

2 How do you choose between the SPM, the SPM_e and the DFN?

By the current. The single particle model holds below about 1C, where the electrolyte stays nearly uniform; the SPM_e adds the electrolyte and holds to about 3C; the DFN resolves the electrode beyond that. A bigger model than the current needs is harder to train and buys nothing. *Ex_10.1 section 3*

3 – 4 · THERMAL COUPLING

3 Why must the thermal and the electrochemical model be solved together?

The reaction and the transport generate heat, and the temperature changes the rates and the diffusivities that generate it, so the loop runs both ways. A decoupled model gets the trend wrong, not only the size. *Gu & Wang (2000)*

4 Why is scaling the first task here, and what does a PINN add when PyBaMM already solves the model?

The raw quantities span twenty orders of magnitude, and no loss weight makes up for that. PyBaMM supplies the parameters and the reference. The network is a differentiable model: it can identify a parameter from a voltage curve and estimate what a battery management system cannot measure. *Ex_10.1 sections 3, 6*

Ex_10.1 — A Particle in a Battery Cell

TASKS

- 1 Sections 1 to 3: a graphite particle in a 1C discharge, its data, its scales and its mass balance.
- 2 Section 4: the exact series, and finite volumes for an agreed 30 mol/m^3 .
- 3 Section 5: a PINN with the start built in and the mass balance in the loss; the three compared.
- 4 Section 6: the diffusivity recovered from 37 noisy readings of the surface concentration.
- 5 Sections 7 to 9: what it says, your report, and mini projects MP10.1A and MP10.1B.

QUESTIONS, AND WHERE ANSWERED

- | | |
|--|-----------|
| Why does the SPM fail as C-rate rises? | sl. 5, 10 |
| <hr/> | |
| Why must the coupling run both ways? | sl. 14 |
| <hr/> | |
| Which heat term changes sign, and why? | sl. 13 |
| <hr/> | |
| Why is scaling harder here than in L8 or L9? | sl. 17 |
| <hr/> | |
| What can be inferred from a voltage curve alone? | sl. 19 |

PyBaMM supplies the parameters and the reference. The PINN supplies the differentiable model — which is what L10.2 then optimises.

PyBaMM Already Solves This — So Why a PINN?

USE PYBAMM

Any forward simulation.
It is faster, validated, and
maintained by the community.
Full DFN with degradation,
ageing and thermal options.
There is no contest here.

USE A PINN

Parameter identification from
measured voltage curves.
Estimating internal states that
no sensor can reach.
A differentiable model that an
optimiser can be run through.

- **say this plainly** — for a forward simulation of a cell, PyBaMM is the better tool
- faster, verified, and maintained by people who do this full time
- the reason to build the PINN is that it is differentiable

The PINN does not compete with the solver: it is a differentiable surrogate constrained by the physics.

L10.2: From a Cell to a Stack

- L10.2 takes the same three physics to a stack
- species transport, charge conservation, and heat
- the equations are recognisably the same
- what changes is the temperature range, and the fact that gradients crack ceramic
- and the channel flow closes the loop with L9

WHAT CARRIES ACROSS

species transport	same operator
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charge conservation	same operator
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heat with a source	same operator
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Butler–Volmer	same form
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operating temperature	600–900 °C
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gas-phase channel flow	new — from L9
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four of six rows are already familiar

Key Takeaways

- 1 Five familiar equations**
Parabolic transport, elliptic charge conservation, parabolic heat. Each one you have solved before; what is new is solving them together.
- 2 Choose the model, not the biggest**
SPM below 1C, SPM_e to about 3C, DFN beyond. Using a DFN where an SPM suffices costs conditioning and buys nothing.
- 3 The coupling runs both ways**
Heat changes the properties that generate the heat. Gu & Wang show a decoupled model gets the trend wrong, not merely the magnitude.
- 4 Scaling is the first task**
Twenty orders of magnitude between raw quantities. No weighting scheme substitutes for non-dimensionalisation.
- 5 The win is what you cannot measure**
Core temperature, internal concentration, state of health. A BMS sees voltage, current and surface temperature — and needs the rest.

Next — L10.2: solid oxide cells, and the real-time optimisation this has all been building towards.

Sources for This Lecture

Gu & Wang (2000)

Thermal-electrochemical coupled modeling of a lithium-ion cell: the heat-generation terms (Eq. 25), the Arrhenius dependence (Eq. 26), coupled against decoupled.

Doyle, Fuller & Newman

The DFN framework: porous-electrode theory, concentrated-solution transport and Butler–Volmer kinetics. Doyle et al. (1996) supplies the OCP expressions used here.

Chen et al. (2020)

Parametrisation of the LG M50 21700 cell: a complete, public DFN parameter set for a real cylindrical cell. PyBaMM ships it as Chen2020.

O'Regan et al. (2022)

The thermal follow-up to Chen2020, supplying the measured thermal and entropic parameters that Chen2020 lacks. PyBaMM ships it as ORegan2022.

PyBaMM

Open-source battery modelling framework. Used here to generate reference solutions and to supply parameter sets — not as a competitor to the PINN but as the trusted comparison.

Parameter sets and reference data are public: every number in the exercise can be traced to a published measurement.